
EE1060: Differential Equations and Transform Techniques

Lecture 21: Fourier Series

Topics :

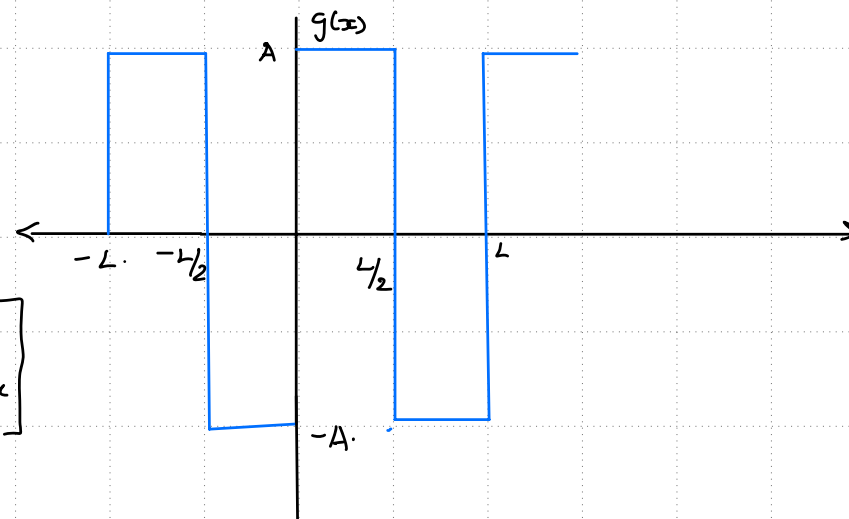
1. Symmetry and its impact on Fourier Coefficients
 2. Effect of time-shift and Derivative on Fourier Coefficients
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Example

$$\omega_0 = \frac{2\pi}{L}$$

$$C_k = \frac{1}{L} \int_0^L g(x) e^{-jk\omega_0 x} dx$$

$$= \frac{1}{L} \left[\int_0^{L/2} A e^{-jk\omega_0 x} dx - \int_{L/2}^L A e^{-jk\omega_0 x} dx \right]$$

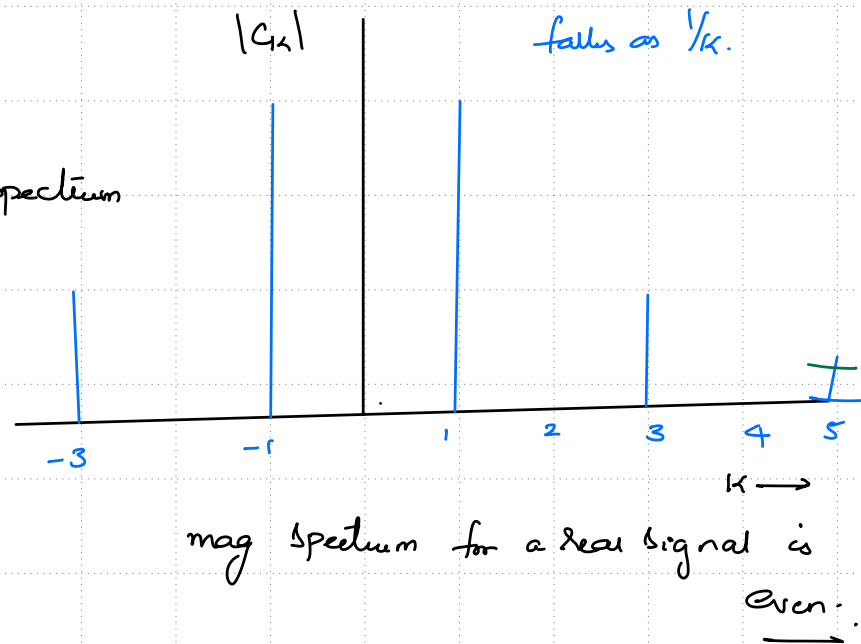


$$C_k = \begin{cases} -j \left(\frac{2A}{k\pi} \right) & \text{when } k = \text{odd} \\ 0 & \text{when } k = \text{even} \end{cases}$$

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magnitude of C_k vs $k \rightarrow$ magnitude spectrum

Phase of C_k vs $k \rightarrow$ phase spectrum



Fourier Coefficients for Even and Odd functions

Even function:- $g(-x) = g(x)$

Trig Fourier Series:
$$g(x) = \underbrace{a_0}_{\text{even}} + \sum_{n=1}^{\infty} \underbrace{a_n \cos n\omega_0 x}_{\text{even}} + \underbrace{b_n \sin n\omega_0 x}_{\substack{\downarrow \\ \text{odd}}}$$

Complex Fourier Series:

$$\begin{aligned}
 c_k &= \frac{1}{L} \int_{-L/2}^{L/2} g(x) e^{-jk\omega_0 x} dx \\
 &= \frac{1}{L} \left[\int_{-L/2}^0 g(x) e^{-jk\omega_0 x} dx + \int_0^{L/2} g(x) e^{-jk\omega_0 x} dx \right] \\
 &\quad u = -x \\
 &= \frac{1}{L} \left[\underbrace{\int_0^{L/2} g(u) e^{+jk\omega_0 u} du}_{A^*} + \underbrace{\int_0^{L/2} g(x) e^{-jk\omega_0 x} dx}_A \right] \\
 c_k &= \frac{1}{L} (A + A^*) \Rightarrow b_k = 0
 \end{aligned}$$

Odd function:- $g(-x) = -g(x)$

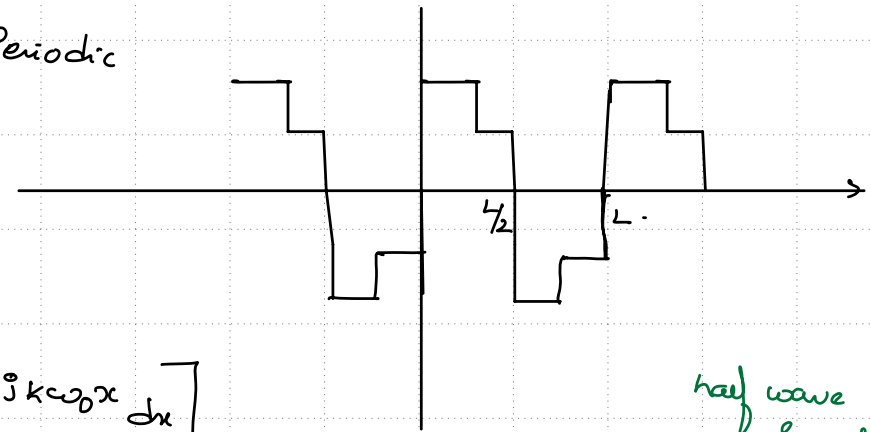
Trig Fourier Series:
$$g(x) = \underbrace{a_0}_{\text{even}} + \sum_{n=1}^{\infty} \underbrace{a_n \cos n\omega_0 x}_{\text{even}} + \underbrace{b_n \sin n\omega_0 x}_{\substack{\downarrow \\ \text{odd}}}$$

$$a_n = 0 \quad \forall n \quad \text{inc } n=0 \Rightarrow a_0 = 0 \Rightarrow \text{DC Component} = 0.$$

Fourier Coefficients for functions with half-wave symmetry

$g(x)$ is Periodic

$$g(x + L/2) = -g(x) \text{ for all } 0 \leq x \leq L/2$$



half wave symmetry

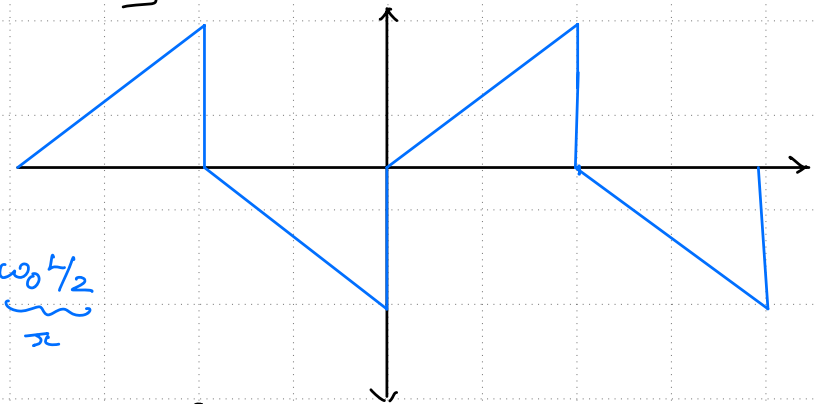
$$C_k = \frac{1}{L} \left[\int_0^{L/2} g(x) e^{-jk\omega_0 x} dx + \int_{L/2}^L g(x) e^{-jk\omega_0 x} dx \right]$$

$$u = x - L/2$$

$$dx = du$$

$$\omega_0 x = \omega_0 u + \omega_0 L/2$$

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$$\int_0^{L/2} \underbrace{g(u + L/2)}_{g(u)} e^{-jk\omega_0 u} \cdot e^{-jk\pi} du$$

$$(-1)^{k+1} \int_0^{L/2} g(u) e^{-jk\omega_0 u} du$$

$$C_k = \begin{cases} 0 & k = \text{even} \\ \frac{2}{L} \int_0^{L/2} g(x) e^{-jk\omega_0 x} dx & k = \text{odd} \end{cases}$$

No even harmonics